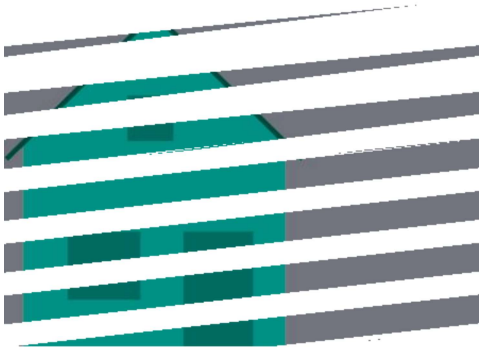
Project 1: Greener Houses (1)

How can EPC and socio-economic data be used to identify which homes in Sheffield most need **energy-efficiency** upgrades and support better place-based decisions?

Project 1: Greener Houses (2)

Context

- Local authorities need better evidence to decide where **energy-efficiency upgrades** are most needed.
- **EPC (Energy Performance Certificate) data** can help identify housing patterns and retrofit needs across Sheffield.



- Socio-economic data can add context on where support may be most needed.
- The project will explore how these insights can inform better local decision-making

Project 1: Greener Houses (3)

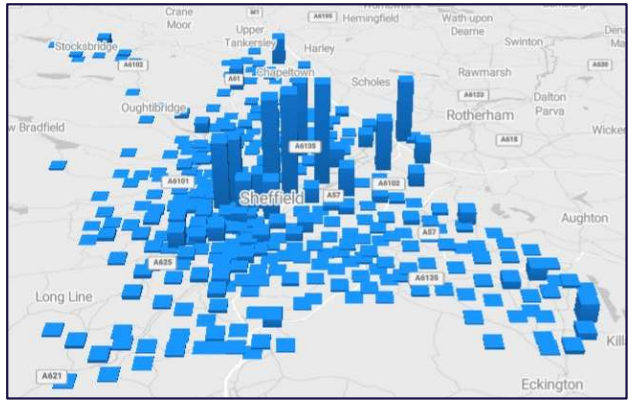
What has been done?

- EPC data has been used, but it has not been widely explored at the local level to develop **housing archetypes** for cities such as Sheffield.
- Energy data has not been widely combined with **socio-economic data** to identify where retrofit needs (insulation and heating upgrades) are greatest.

Project 1: Greener Houses (3)

Questions:

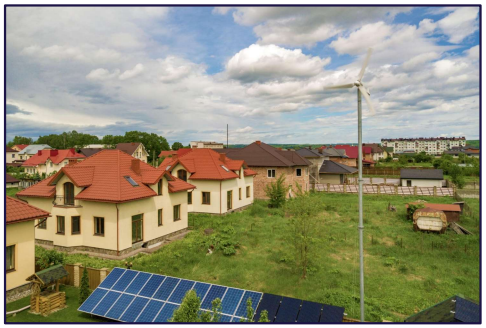
1. How can EPC data help identify different **housing archetypes** in Sheffield?
2. Which **areas** show the greatest need for energy-efficiency upgrades?
3. How can socio-economic data help prioritise where support may be most needed?
4. How can these insights be communicated clearly through maps, dashboards, or other visual designs?



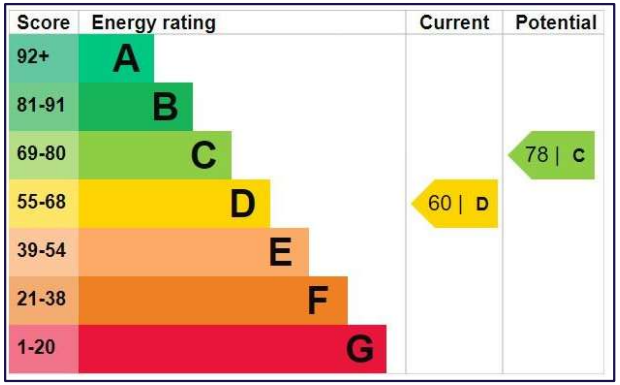
Matrix of Correlations	Average download speed (Mbit/s)	Average Weekly Earnings (Pounds)	Business Churn Rate per 100,000 inhabitants	Collected household waste per person (kg)	Deaths of Business	Deaths under 75 from preventable causes (per 100,000)	Employment rate	Female healthy life expectancy (years)	Female life expectancy (years)	Gross Value Added in Millions of Pounds	Gross weekly pay (Pounds) for the Highest Paid 80 Percent	Gross weekly pay (Pounds) for the Lowest Paid 10 Percent	Inequ 80th 10 Wee
Average download speed (Mbit/s)	1.00	0.69	0.22	-0.07	-0.16	0.26	-0.06	-0.34	-0.20	-0.02	0.64	0.60	
Average Weekly Earnings (Pounds)	0.69	1.00	-0.03	-0.24	0.16	0.21	0.40	-0.14	-0.04	0.37	0.95	0.90	
Business Churn Rate per 100,000 inhabitants	0.22	-0.03	1.00	0.35	-0.85	0.27	-0.19	-0.47	-0.34	-0.87	-0.02	0.05	
Collected household waste per person (kg)	-0.07	-0.24	0.35	1.00	-0.32	-0.23	0.26	0.11	0.16	-0.32	-0.17	-0.19	
Deaths of Business	-0.16	0.16	-0.85	-0.32	1.00	-0.18	0.12	0.39	0.29	0.88	0.08	0.04	
Deaths under 75 from preventable causes (per 100,000)	0.26	0.21	0.27	-0.23	-0.18	1.00	-0.42	-0.71	-0.82	-0.22	0.15	0.34	
Employment rate	-0.06	0.40	-0.19	0.26	0.12	-0.42	1.00	0.54	0.53	0.25	0.46	0.23	
Female healthy life expectancy (years)	-0.34	-0.14	-0.47	0.11	0.39	-0.71	0.54	1.00	0.82	0.40	-0.06	-0.37	
Female life expectancy (years)	-0.20	-0.04	-0.34	0.16	0.29	-0.82	0.53	0.82	1.00	0.35	0.02	-0.22	
Gross Value Added in Millions of Pounds	-0.02	0.37	-0.87	-0.32	0.84	-0.22	0.25	0.40	0.35	1.00	0.28	0.25	
Gross weekly pay (Pounds) for the Highest Paid 80 Percent	0.64	0.95	-0.02	-0.17	0.08	0.15	0.46	-0.06	0.02	0.28	1.00	0.86	
Gross weekly pay (Pounds) for the Lowest Paid 10 Percent	0.60	0.90	0.05	-0.19	0.04	0.34	0.23	-0.37	-0.22	0.25	0.86	1.00	
Inequality: Ratio of	-0.29	-0.66	-0.13	0.13	0.06	-0.42	0.01	0.52	0.38	-0.13	-0.56	-0.89	

Project 1: Greener Houses (4)

Design perspective: **Dashboard / Report + House Design**



Project 1: Greener Houses (5)

Let's talk data!

- Main dataset: https://epc.opendatacommunities.org/
  - 1. Register.
  - 2. Aggregate the data by house archetypes or similar (**privacy issues**).
- Socioeconomic data: https://www.nomisweb.co.uk/
  - 1. Go to "Query Data".
  - 2. Select the indicators you are interested (Earnings, Education, etc.).
- Alternatively: Explore the Index for Multiple Deprivation (IMD)